# 8.3 Box-and-Whisker Plots

Name _____ Period ____ Date _____

Algebra I 8.3 · Student sheet

**I can…**

- find the five-number summary of a data set.
- make and read a box-and-whisker plot.
- use the range, interquartile range, and shape to describe and compare data sets.

**Vocabulary** · fill in the missing word

**Quartiles** — values that _____ a data set into four equal parts: Q1 is the median of the lower half, Q3 the median of the upper half

**Five-number summary** — least value, Q1, median, Q3, _____ value

**Box-and-whisker plot** — a display of the _____ summary; the box goes from Q1 to Q3 with a line at the median, and the whiskers reach the least and greatest values

**Interquartile range (IQR)** — Q3 − Q1, the _____ of the middle half of the data

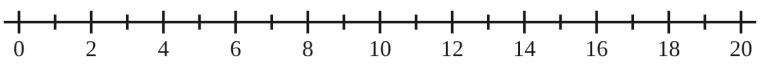
## Part 1 · Human box plot

Record the numbers on the cards in order:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | |

| Least | Q1 | Median | Q3 | Greatest | Range | IQR |
|---|---|---|---|---|---|---|
| | | | | | | |

Sketch the box-and-whisker plot:




## Part 2 · Notes: fill in the blanks

**Five-number summary:** _____, _____, _____, _____, _____

Q1 is the median of the _____ half; Q3 is the median of the _____ half.

**IQR** = _____ − _____, the spread of the middle _____ of the data.

A longer right whisker means the distribution is skewed _____.

## Part 3 · Examples (work them with the class)

### Example 1

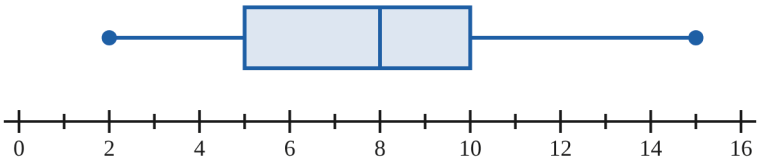
Ages of 10 people at a party: 18, 25, 21, 30, 19, 23, 27, 22, 20, 35. Find the five-number summary and make a box-and-whisker plot.

**Example 2**

Use the box-and-whisker plot.

a. Find the median. b. Find the range. c. Find the IQR.

d. About what percent of the data are between 5 and 10?

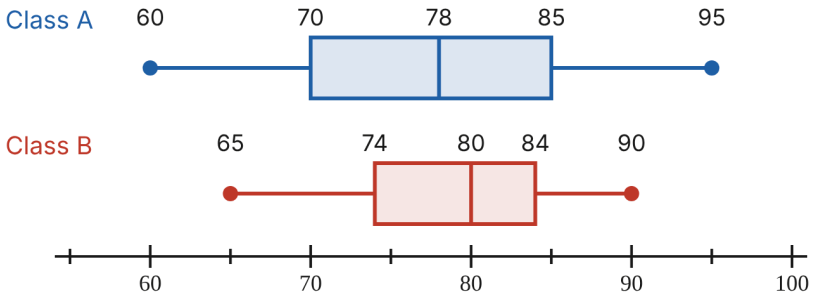



**Example 3**

The plots show the quiz scores of two classes.

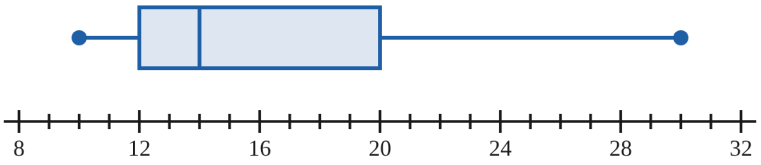
a. Which class has the greater median?

b. Which class's scores are more spread out? Use the range and IQR.



### Example 4

Describe the shape of the distribution.




## Part 4 · Partner practice

### Partner 1

Find the five-number summary and the IQR: 7, 15, 9, 12, 20, 11, 14, 8, 16

### Partner 2

Find the range and the IQR. Are the data more spread out below Q1 or above Q3?

